

Date: 5.5.2025

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Lab 3: Heat

Overview

In Lab 4, we measured the transient temperatures of a stainless-steel plate and a copper sheet after they were heated to 50 °C, and found the steady-state temperature along three metal fins (copper, aluminum, and stainless steel). A thermal heat gun recorded the sheet temperatures, while thermocouples spaced at known intervals from each fin's base were fed into a selector-dial digital readout. By fitting these temperature profiles to the appropriate analytical solutions, we found the convective heat-transfer coefficient, h , for both experiments and calculated the Biot number.

Setup

Figure 1 shows transient measurements:

- Two metal samples (a 304 stainless steel plate and a pure copper sheet) were clamped together, with TC 20 measuring the steel side.
- A heat gun (200 °C) was directed perpendicularly at 6 inches from the steel surface until the TC20 reading reached 50 °C.
- At 50 °C, the gun was turned off, and the temperature was recorded every 10 s until the sample returned to ambient.
- The steel plate was then flipped, and the procedure was repeated 3 times for the copper sheet.

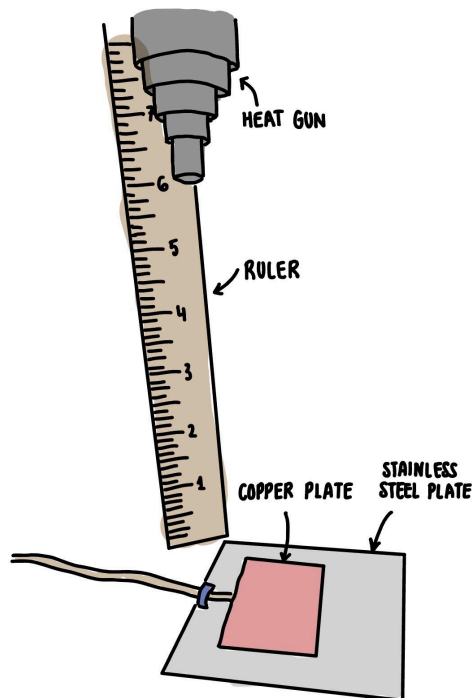


Figure 1: Copper plate (top) and stainless steel plate (bottom) configuration with thermocouple connection. A heat gun is positioned 6 inches above the plate to heat the sample.

Figure 2 shows Steady-State Fin Measurements:

- Three cylindrical fins (copper, aluminum, stainless steel), each 9.55 mm in diameter, were attached to a 500 °C hot plate and allowed to reach steady state.
- Thermocouples (TC 1–15) measured the fin temperatures at zip-tie distances of 30, 60, 120, 240, and 455 mm from the base.
- An IR thermometer provided an independent temperature measurement at the same five locations.

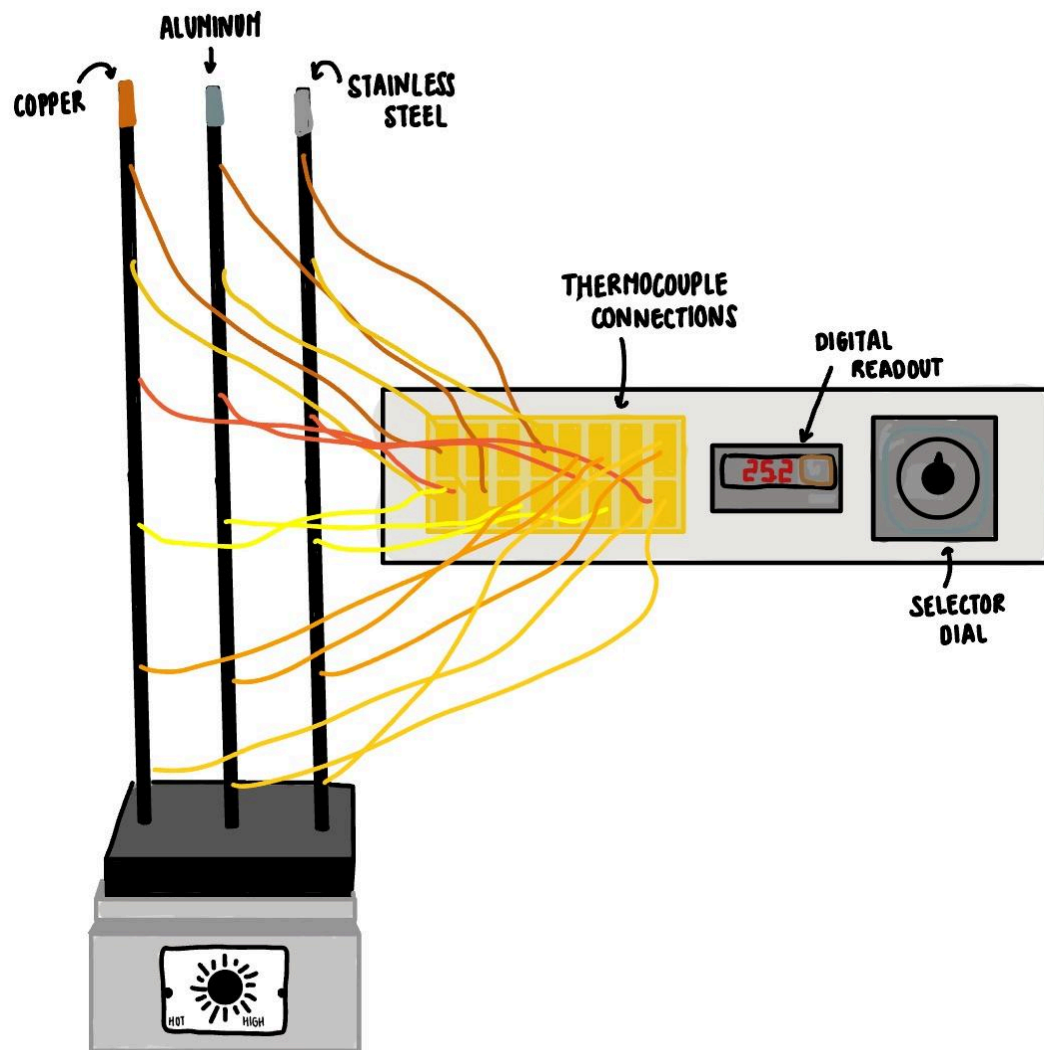


Figure 2: Copper, stainless steel, and aluminum heat fins are attached to thermocouples. The digital readout shows the temperature.

Transient Measurements Analysis

Table 1. Material Properties

	Copper sheet (pure)	Stainless Steel plate (304)
C _p [J/gC]	0.385	0.500
k [W/mK]	401	16.2
A [cm ²]	15.75	59.29
V [cm ³]	0.4725	11.858
ρ [g/cm ³]	8.96	8.00

Figure 1: Material properties of copper sheet and stainless steel

TC Measurements (#'s)	Copper (1-5)	Aluminum (6-10)	Stainless Steel (11-15)
30mm (1,6,11)	132 °C	111 °C	92.8 °C
60mm (2,7,12)	117.2 °C	85.6 °C	63.5 °C
120mm (3,8,13)	93.2 °C	82.8 °C	38.8 °C
240mm (4,9,14)	65.3 °C	51 °C	26.7 °C
455mm (5,10,15)	47.7 °C	34.5 °C	24.2 °C

Figure 2 shows thermocouple readings for TCs 1-15

IR Measurements	Copper	Aluminum	Stainless Steel
(Closest) 30mm	130 °C	147 °C	133 °C
60mm	101 °C	118 °C	83 °C
120mm	79 °C	70 °C	39 °C
240mm	53 °C	45 °C	24.2 °C
(Farthest) 455mm	37 °C	29.5 °C	21.7 °C

Figure 3 shows the temperatures registered from the IR thermometer

Time vs Temperature plots for Stainless Steel and Copper Sheet

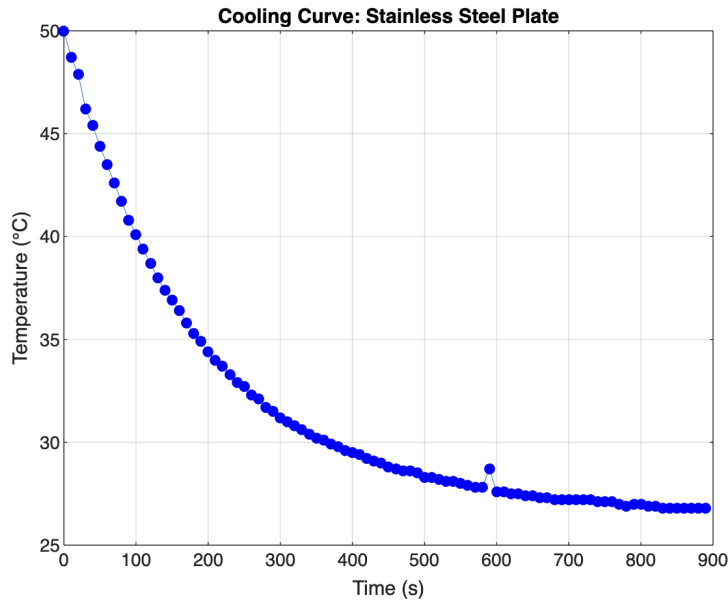


Figure 4: Time-temperature profile of stainless steel during cooling. The blue circles mark the steel-plate temperature every 10 s as it falls from 50 °C at $t = 0$ down toward the ambient level (25 °C). The curve is roughly exponential: the plate cools from 50 °C to about 30 °C in the first 300 s, then slowly drifts the last few degrees over the next 600 s.

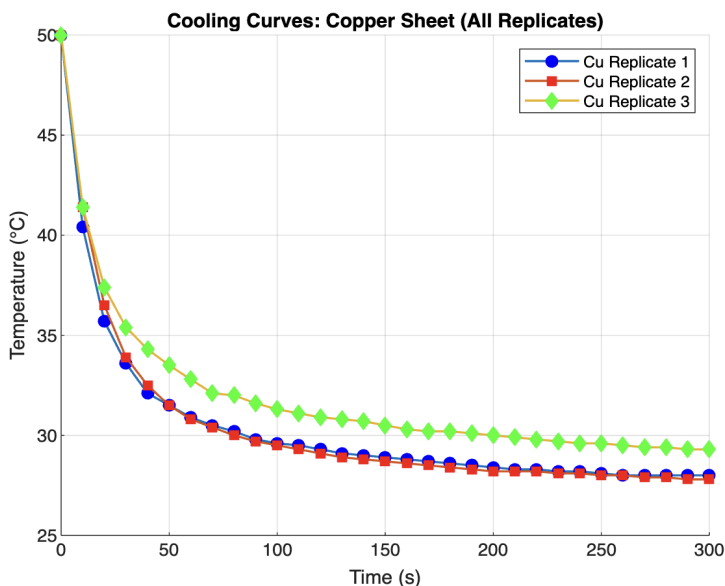


Figure 5: Time-temperature profile of copper during rapid cooling. Three copper-sheet replicates are plotted. Replicate 1 (blue circles), replicate 2 (red squares), and replicate 3 (green diamonds). Replicates 1 and 2 coincide almost exactly, while replicate 3 remains about 1–2 °C warmer throughout, likely due to small alignment or sensor-placement differences. All three drops from 50 °C to around 30 °C by around 100 s, then level off near ambient by about 300 s, illustrating both copper's faster heat conduction (steeper initial slope) and the excellent repeatability of the measurements.

Calculating the Convective Heat Transfer to Air (h) for each Stainless Steel and Copper

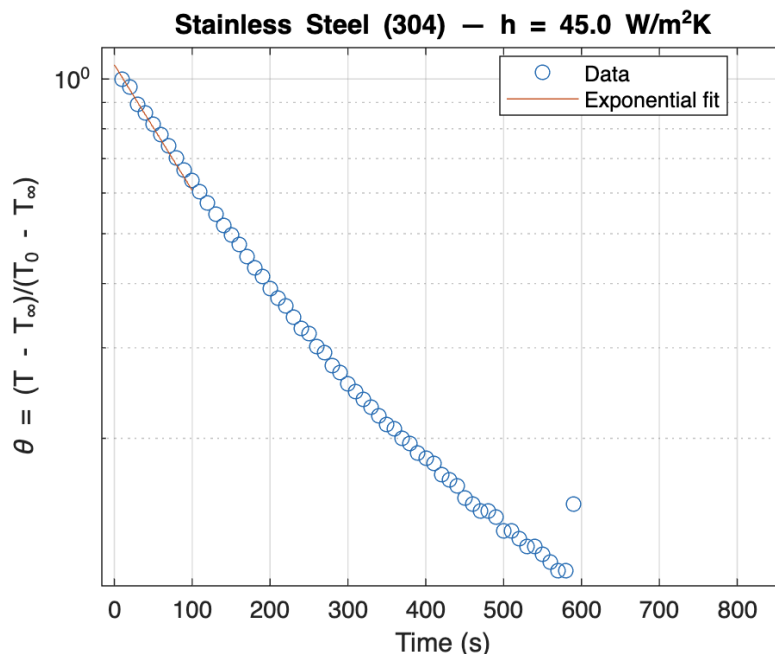


Figure 6: This graph shows how the normalized temperature falls over time as the 304 stainless-steel plate cools by unsteady-state conduction to an infinite air sink. The vertical axis (θ) is on a logarithmic scale, while time is on a linear scale. Open circles mark the measured temperature transients, and the solid orange line is an exponential fit applied to the initial straight-line portion of the curve (first 100 s). The slope of that fit yields the convective heat-transfer coefficient, $h = 45.5 \text{ W/m}^2\cdot\text{K}$.

$$\frac{T - T_{\infty}}{T_0 - T_{\infty}} = e^{-\frac{hAt}{\rho c_p V}}$$

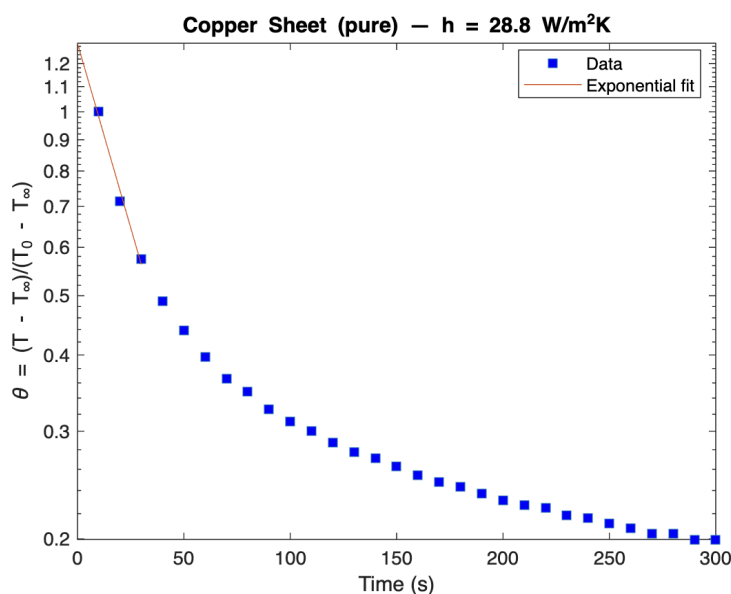


Figure 7: This plot presents the normalized temperature versus time for the pure copper sheet cooling under the same conditions. The temperature ratio axis is logarithmic, and time is linear. Filled squares are the experimental data points, and the solid orange line is the exponential fit applied to the early linear segment (first 30 s) of the decay. From its slope we determine the copper's convective coefficient, $h = 28.8 \text{ W/m}^2 \cdot \text{K}$.

Calculating the Biot Modulus for Copper Sheet and Steel Plate

Using Biot Modulus using the equation $Bi = hV/Ak$, with our measured h and the known k , V , and A for each plate, we find Biot numbers well below 0.1 for both copper and stainless steel. These low Biot numbers show that heat moves through the metal much faster than it leaves its surface. That means it's reasonable to treat each plate as having one uniform temperature during the cooling tests. As we expected, copper's Biot number is even smaller than stainless steel's because copper conducts heat much better. These hand-calculated Biot numbers therefore support the accuracy of our experimental results.

Calculating Biot Number

Copper: $Bi_{\text{copper}} = \left(\frac{hV}{Ak} \right)_{\text{copper}} = \frac{(28.8 \text{ W/m}^2 \cdot \text{K}) (0.4725 \times 10^{-5} \text{ m}^3)}{(15.75 \times 10^{-3} \text{ m}^2) (401 \text{ W/m} \cdot \text{K})} = 2.15 \times 10^{-5}$

Stainless steel: $Bi_{\text{stainless steel}} = \left(\frac{hV}{Ak} \right)_{\text{stainless steel}} = \frac{(45) (11.858 \times 10^{-5})}{(59.29 \times 10^{-3}) (16.2)} = 5.55 \times 10^{-3}$

Figure 8: Shows the Biot Modulus number calculated by hand.

Steady- State Measurements

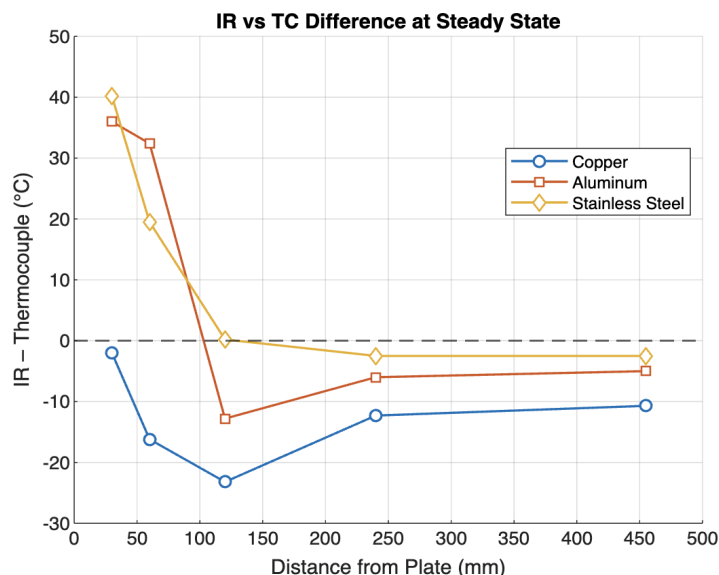


Figure 9: This plot shows how the IR thermometer's reading minus the thermocouple's reading varies with distance along the fin. Positive values mean the IR device reads hotter. You can see that as you get closer to the heat source, the IR–TC gap grows, showing a consistent overestimation by the IR thermometer at shorter distances. This pattern shows a steady error, probably because the fin gives off heat differently near the hot end. Small random mistakes like the zip-tie markers shifting or the thermometer being aimed slightly off can also affect the readings.

Temperature of a Fin with the Use of the IR Thermometer Data

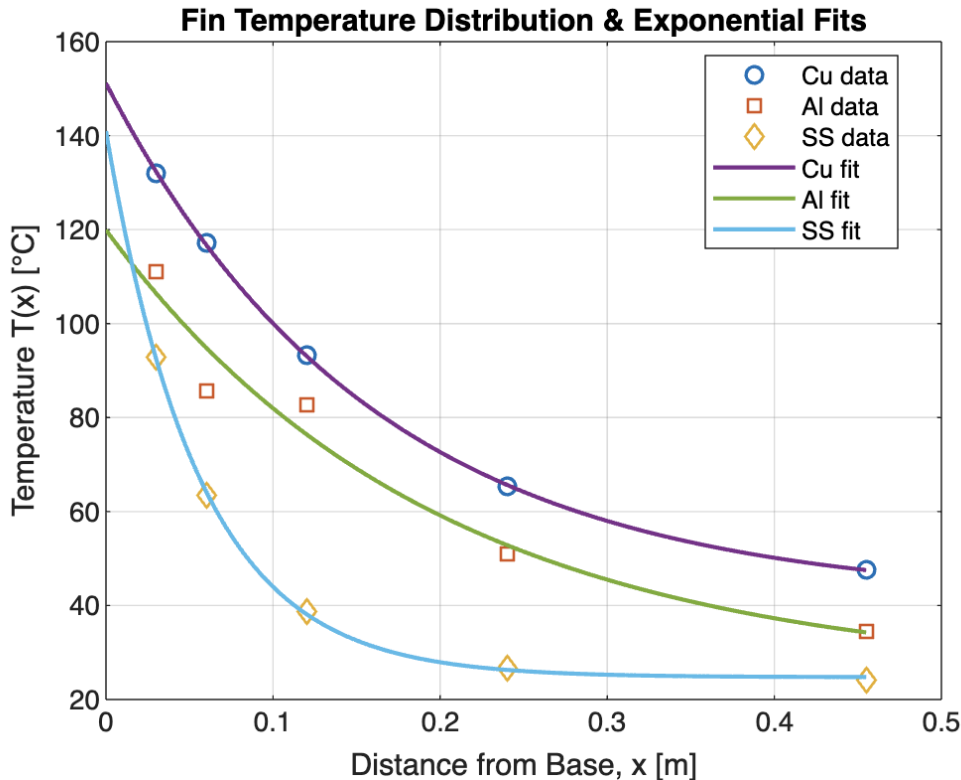


Figure 10: This plot shows the IR-thermometer measurements of fin temperature (blue circles for copper, orange squares for aluminum, yellow diamonds for stainless steel) as you move away from the heated base. Each set of IR readings falls on a smooth, exponentially decaying curve that levels off near the ambient temperature (25 °C) at the fin tip. Stainless steel cools fastest. The IR data drops from about 95 °C at the root to nearly ambient by 0.24 m. Aluminum instead sits in the middle and copper cools most slowly, still around 50 °C at 0.455 m. The solid exponential fits (purple for copper, green for aluminum, light blue for stainless steel) align almost perfectly with the IR points, confirming that the classical fin equation $\frac{dx^2}{dT} - hPkA(T - T_{inf}) = 0$, accurately predicts the temperature distribution under natural convection.

Computed convective coefficients:
 Copper fin: $h = 37.4 \text{ W/(m}^2\cdot\text{K)}$
 Aluminum fin: $h = 12.8 \text{ W/(m}^2\cdot\text{K)}$
 Stainless Steel: $h = 12.4 \text{ W/(m}^2\cdot\text{K)}$

Discussion

In this lab, we studied heat transfer behavior through transient cooling and steady-state fin measurements. We found that copper cooled faster than stainless steel because it carries heat more easily. We measured heat-transfer rates of $28.8 \text{ W/m}^2\cdot\text{K}$ for copper and $45 \text{ W/m}^2\cdot\text{K}$ for steel. The low Biot numbers mean heat moved inside the metal much quicker than it left the surface, so we could treat each piece as having one uniform temperature. In the fin tests, temperature dropped along the fin, giving heat-transfer rates of $37.4 \text{ W/m}^2\cdot\text{K}$ for copper fins, $12.8 \text{ W/m}^2\cdot\text{K}$ for aluminum fins, and $12.4 \text{ W/m}^2\cdot\text{K}$ for steel fins. We also saw that the IR thermometer read too high near the fin bases, probably because of surface emissivity, while the thermocouple readings matched what we expected.